

Data 100, Summer 2026

Discussion #4 Solutions

Note: Your TA will probably not cover all the problems. This is totally fine; the discussion worksheets are not designed to be finished within an hour. They are deliberately made slightly longer so they can serve as resources you can use to practice, reinforce, and build upon concepts discussed in lectures, labs, and homework.

1 variable	Type of visualization	Description
Quantitative	Histogram Boxplot Density plot (KDE)	Distribution (shape/center/spread/skew) Summary statistics (median/IQR/outliers) Smoothed distribution
Qualitative	Bar chart	Counts/frequency of each category

2 variables	Type of visualization	Description
Two quantitative	Scatter Hex Contour Line	Relationship (correlation/trends/outliers) Reduces overplotting Density/level sets (2D concentration) Temporal trends over time
Quantitative and qualitative	Boxplot per category Violin per category	Summary statistics per category (median/IQR/outliers) Shape and distribution per category
Two qualitative	Stacked bar chart	Part-to-whole within categories

3 and more variables	Type of visualization	Description
All quantitative	Heatmap (e.g. correlation matrix)	Strength and direction of correlations
Mixed	Scatter with color, shape, size	Encodes additional variables via visual channels

1. Bigfoot is a mysterious ape-like creature that is said to live in North American forests. Most doubt its existence, but a passionate few swear that Bigfoot is real. In this question, we will be using a DataFrame `bigfoot`. The `bigfoot` data contains many variables about each reported Bigfoot spotting, including location information, weather, and moon phase. Here are the first few entries of `bigfoot`:

season	date	temperature_high	temperature_low	humidity	cloud_cover	moon_phase	precip_intensity	pressure	uv_index	visibility	wind_speed
Summer	2016-06-07	74.69	53.80	0.79	0.61	0.10	0.0010	998.87	6.0	9.70	0.49
Summer	2015-10-02	49.06	44.24	0.87	0.93	0.67	0.0092	1022.92	3.0	9.16	2.87
Fall	2009-10-31	69.01	34.42	0.77	0.81	0.42	0.0158	1011.48	3.0	1.97	3.94
Summer	1978-07-15	68.56	63.05	0.88	0.80	0.33	0.0285	1014.70	5.0	5.71	5.47
Summer	2015-11-26	20.49	5.35	0.65	0.08	0.54	0.0002	1037.98	1.0	10.00	0.40

Note: The original dataset is extremely messy; for the purposes of this discussion, we will drop all observations with any missing values and some unneeded columns.

- (a) Rohan is very interested in Bigfoot, and he wants to know the typical wind speeds when Bigfoot sightings are reported. Which of the following are appropriate visualizations for plotting the distribution of `wind_speed`?

- ☐ A. Pie chart
- ☒ B. Kernel Density Estimation plot
- ☐ C. Scatter plot
- ☒ D. Box plot
- ☒ E. Histogram
- ☐ F. Hex plot

Solution: `wind_speed` is a single quantitative variable. This rules out pie charts, which are used to visualize qualitative variables. It also rules out scatter plots and hex plots, which require at least two quantitative variables. Therefore, Kernel Density Estimation plots, box plots, and histograms are appropriate visualizations for the distribution of `wind_speed`.

- (b) Rohan then wants to examine which season Bigfoot is spotted in most often. Which visualization would be appropriate for showing the number of sightings in each season?

- ☐ A. Heatmap
- ☐ B. Box plot
- ☒ C. Bar chart
- ☐ D. Kernel Density Estimation plot
- ☐ E. Line plot

Solution: `season` is a qualitative variable, and Rohan wants to count how many sightings occur in each category. A bar chart is appropriate for visualizing the frequency of each category. Heatmaps usually require two variables, box plots and Kernel Density Estimation plots are used for quantitative variables, and line plots are typically used for ordered or time-series data. Therefore, a bar chart is the most appropriate choice.

- (c) As Rohan explores this dataset, he is interested in understanding the relationship between `windspeed` and `visibility`. What visualizations would help Rohan understand this relationship?

- ☒ A. Scatter plot
- ☐ B. Box plot
- ☒ C. Contour plot
- ☐ D. Violin plot
- ☐ E. Stacked bar chart

Solution: Both `windspeed` and `visibility` are quantitative variables. Box plots, violin plots, and stacked bar charts all require at least one qualitative variable, while scatter plots and contour plots can show the relationship between two quantitative variables.

Data Collection through Sampling

2. It's time for the Data 100 midterm, and the professors want to estimate the difficulty of the exam. They decided to survey students on the exam's difficulty with a 10-point scale and then use the mean of the student's responses as the estimate.

(a) What is the population the professors are interested in trying to understand?

- ☐ A. Students in Data 100
- ☐ B. Students enrolled in the Data 100 Ed
- ☐ C. Students who attend the Data 100 lectures
- ☒ D. Students who took the Data 100 midterm

Solution: Some students enrolled in Data 100, on Ed, or who attend lectures may not have taken the midterm exam. We only want to survey students who *actually* took the exam, so option D is the correct choice.

- (b) The professors consider a few different methods for collecting the survey data about the midterm. Which of the following methods is best? (Think through which considerations go into "best".)

- ☐ A. The professors add one Slido poll in the first lecture following the exam and only consider synchronous responses.
- ☒ B. The professors add a question to the homework assignments of a simple random sample of students in the entire course.
- ☐ C. The professors make a post on Ed asking students to submit a Google Form containing the survey question.
- ☐ D. The professors choose a simple random sample of discussion sections, go to each selected section and ask each student in the group as part of the final discussion question.

Solution:

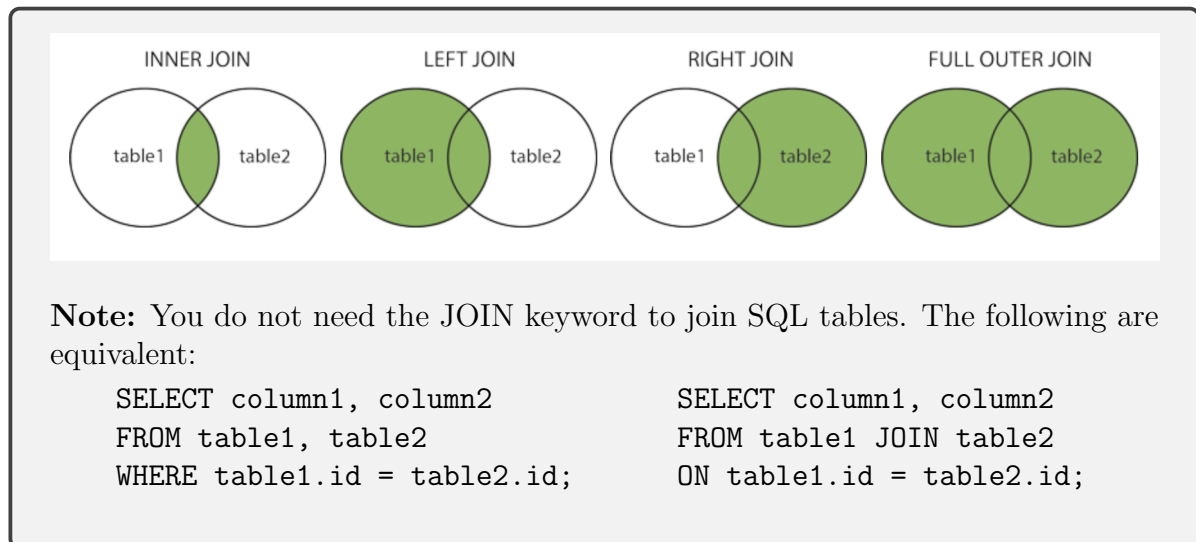
Option A: Although convenient, by sampling only from students who attend synchronously, the professors are restricting students who prefer not to attend live sessions. These can introduce *selection bias* in the sample.

Option B: This method of sampling works well for this scenario because the professors are sampling randomly from students across the entire course, which takes into account many of the axes along which students can differ. They are also adding the question in such a way that students are required to answer it, reducing *non-response bias*, and can answer it privately, reducing social pressure and *reporting bias*.

Option C: As with option A, there are some students who don't look at Ed or ignore certain posts, which introduces *selection bias*. Responding to the survey is also optional and easy to forget, introducing *non-response bias*.

Option D: The primary issue here is one of social pressure because students are being asked in groups, and this can result in *reporting bias*. Additionally, discussion attendance isn't mandatory, there's *selection bias* based on who chooses to attend.

Joins



3. In the figure above, assume **table1** has m records, while **table2** has n records. Describe which records are returned from each type of join. What is the **maximum** possible number of records returned in each join?

Consider the cases where on the joined field, (1) both tables have unique values, and (2) both tables have duplicated values. Finally, what is the **minimum** possible number of records returned in each join?

Solution: (INNER) JOIN: Returns records that have matching values in both tables. The maximum number of rows is $\min(m, n)$ if unique rows in each table. The minimum number of rows is 0.

LEFT (OUTER) JOIN: Return all records from the left table, and the matched records from the right table. The maximum number of rows is m if unique rows. The minimum number of rows is m .

RIGHT (OUTER) JOIN: Return all records from the right table, and the matched records from the left table. The maximum number of rows is n if unique rows. The minimum number of rows is n .

FULL (OUTER) JOIN: Return all records from both tables. If there is a match it will match those rows instead of having both the left and right version of that row. when there is a match in either the left or right table. The maximum number of rows in $m + n$ if unique rows. The minimum number of rows is $\max(m, n)$.

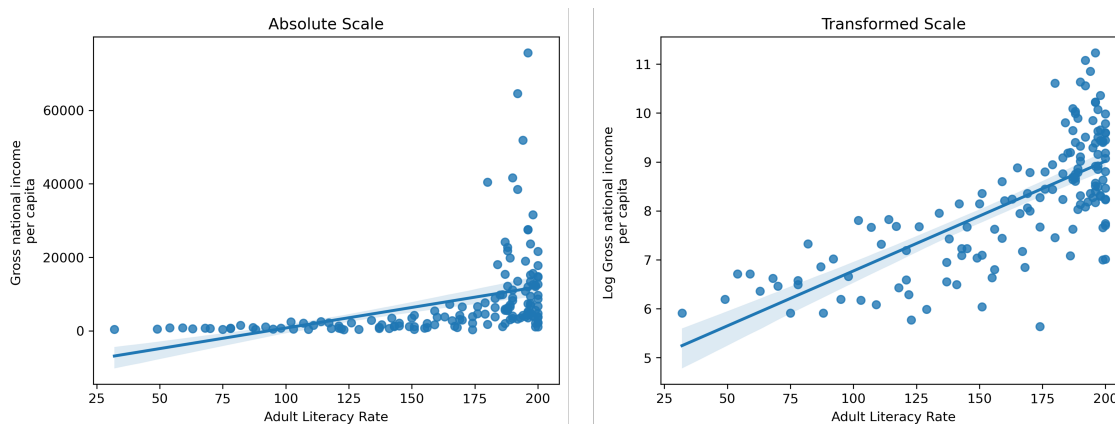
CROSS JOIN: Return all pairs of records between the left and right tables. The maximum number of rows is $m \times n$. The minimum number of rows is $m \times n$.

All joins have a maximum number of $m \times n$ rows if duplicates are allowed.

Logarithmic Transformations

4. Ishani is a development economist interested in studying the relationship between literacy rates and gross national income in countries across the world. Originally, she plotted the data on a linear (absolute) scale, shown on the left. She noticed that the non-linear relationship between the variables with a lot of points clustered towards the larger values of literacy rate, so she decides to do a $\log_e$ transformation of the y-axis, shown on the right.

The solid blue line is a line of best fit (we'll formalize this later in the course).



- (a) Instead of using the $\log_{10}$ transformation of the y-axis, what other transformations could Ishani have used to attempt to linearize the relationship between literacy rate (x) and gross national income per capita (y). Select all that apply.

- A. $\log_{10}(y)$
- B. 10^y
- C. $\sqrt{x}$
- D. x^2
- E. y^2

Solution: The x values are left/negatively skewed. We need to stretch them. The y values are right/positively skewed. We need to squish them. Other possible transformations are: x^3 , $\sqrt{y}$. Note that the base of the logarithm is not specified generally. Both base 10 or base e could work (or most other bases for that matter) in achieving the desired outcome of linearizing the plot, you may decide to use a different base depending on how skewed the data is.

- (b) Let C and k be some constant values and x and y represent literacy rate and gross national income per capita, respectively. Based on the plots, which of the following best describes the pattern seen in the data?

- ☐ A. $y = C + kx$ ☒ B. $y = C \times e^{kx}$ ☐ C. $y = C + k \log_e(x)$ ☐ D. $y = Cx^k$

Solution: $y = kx + b$ is the basic format of the regression line. Since the graph on the right takes the log scale of the y-axis, we can write the equation as $\log_{10} y = kx + b$ and manipulate it until we get y on its own:

$$\log_e(y) = kx + b \quad (1)$$

$$e^{\log_e(y)} = e^{kx+b} \quad (2)$$

$$y = e^{kx} \cdot e^b \quad (3)$$

$$y = e^b \cdot e^{kx} \quad (4)$$

Hence, this equation format fits choice B, $y = C \times e^{kx}$, where $C = e^b$.

- (c) What parts of the plots could you use to make initial guesses on C and k ?

Solution:

- C : e^b , where b is the y-intercept of the solid blue line in the transformed plot.
- k : the slope of the solid blue line in the transformed plot.

- (d) Ishani's friend, Yash, points to the solid line on the transformed plot and says: "Since this line is going up and to the right, we can say that, in general, the higher the literacy rate, the greater the gross national income per capita". Is this a reasonable interpretation of the plot?

Solution: Yes, the observation is equivalent to saying that the slope is positive, which means increases in x correspond to increases in y . This does not mean higher literacy rates *cause* higher gross national incomes, just that they are positively correlated.

- (e) Suppose that instead of plotting positive quantities, our data contained some zero and negative values. How can we reasonably apply a logarithmic transform to this data?

Solution: Logs cannot take on zero or negative values, so we must shift our data into positive quantities before applying a logarithmic transform. To achieve this, we add the magnitude (or absolute value) of the most negative number in our data and then add a small positive number (e.g., 1). As a concrete example,

suppose we had the dataset: $-3, -2, 4$. The most negative number is -3 , so we add $|-3| = 3$ to all numbers to obtain: $0, 1, 7$. This fixes the issue with negative values, but this dataset still contains the value 0 . Hence, we add a small positive number such as 1 , to get the numbers $1, 2, 8$ to which we can finally apply the log operation on.